# Monika Santra

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## Research Focus

**Software & Systems Security with augmented AI/ML** — binary analysis, vulnerability discovery, control-flow and memory safety, malware analysis, and agentic AI security; building scalable, reproducible security tooling.

## Education

- **The Pennsylvania State University** — PA, USA
  *Ph.D. in Computer Science & Engineering; GPA: 4/4* — *2022 - present*
  ***Dissertation:*** *AI-Augmented Static Analysis: Bridging Heuristics and Completeness for Practical Reverse Engineering*
  **Advisors:** Dr. G. Gary Tan
- **Indian Institute of Technology Roorkee** — Uttarakhand, India
  *M. Tech in Computer Science & Engineering; GPA: 9.765/10* — *2015 - 2017*
  ***Dissertation:*** *Design and Implementation of Remote Attestation protocols for servers.*
  **Advisors:** Dr. Sateesh K. Peddoju (IIT Roorkee) & Dr. Anup K. Bhattacharjee (BARC)
- **Govt. College of Engineering & Textile Technology, Serampore (WBUT)** — West Bengal, India
  *B. Tech in Computer Science & Engineering; GPA: 8.86/10* — *2010 - 2014*

## Experience

**The Pennsylvania State University** — PA, USA
*Ph.D. Candidate* — *2022 - present*

- **Research Experience — Binary Analysis & AI/ML for RE**
  **DISA — AI/ML-assisted memory modeling for program analysis**
    - Transformer-based disassembly framework that recovers memory block boundaries on stripped binaries, feeding a Value Set Analysis-driven CFG recovery pipeline for more precise indirect-call resolution.

  **iResolveX — A customizable meta-CFG for indirect-call resolution**
    - Introduces *p-IndirectCFG*, a new CFG construct that lets downstream analyses tune indirect-call pruning (precision vs. recall) to their own needs, rather than assuming one CFG fits every task. *(Under review, NDSS)*

  **CAIR — Separating weak and strong program-analysis evidence for indirect-call resolution**
    - GNN-based refinement over Structured Call Refinement Graphs (SCRGs) that distinguishes weak from strong static evidence, using ML support to resolve indirect calls more precisely; evaluated via AICT reduction on SPEC CPU benchmarks against CupidCall, AttnCall, BinDSA, and Callee.

  **Agentic Fuzzing for Indirect-Call Ground Truth Completeness**
    - Building an AFL++-based agentic fuzzer to establish quantifiably complete ground truth for indirect-call target pairs via a residual-risk framing across confirmed, proven-infeasible, and unresolved pairs.

**IBM Thomas J. Watson Research Center** — Yorktown Heights, NY
*Research Scientist Intern* — *May 2026 - Aug 2026*

- **Research Experience — Agentic AI Security ACaM (Agent Capability Model) — Neurosymbolic framework for agentic vulnerability discovery**
    - Combines static analysis with guarded LLM semantic lifting to build the Agent Capability Property Graph, a compositional model of an agent's true capability surface, used to reconstruct multi-hop threat chains and surface structural vulnerabilities across a 43-agent real-world corpus.

**Bhabha Atomic Research Centre (BARC)** — Maharashtra, India
*Research Scientist, Grade D* — *2019 - 2022*

- **Research Experience — Software Security & ML**
    - **Automated Multilayered Malware Classifier**: Designed, built, and deployed an ensemble malware classifier combining basic/advanced static and dynamic analysis; introduced CFG-driven shallow+deep features; integrated with an Application Whitelisting framework for host-side protection.
    - **Binary Vulnerability Detection via LLVM IR**: Developed a static analysis-driven detector for high-frequency issues (e.g., buffer overflows, memory leaks) in third-party binaries prior to deployment; contributed LLVM IR-level data flow-analysis checks.
    - **SPERTS — Safe Programming Environment for Real-Time Systems**: Developed a Linux-based graphical modeling framework and simulation support for safety-critical I&C systems; implemented complex built-in operators and automated LaTeX/JavaScript report generation.

*Research Scientist, Grade C* *2017 - 2019*

- **Research Experience — Systems & Network Security**
  - **Trusted Systems for Instrumentation & Control (I&C) Network**: Designed and demonstrated the in-house Hardware Reference Monitor (PCIe & USB): delivered **Secure Boot**, **Authentication**, and **Remote Attestation** for host/guest Linux and Windows; built kernel/device drivers & daemons, Application Whitelisting, and a real-time trust dashboard; extended QEMU-KVM for virtualization support; contributed SRAM characterization for PUF-based hardware security.
  - **Advanced Industrial Control Firewall (Modbus/TCP)**: Implemented a DPI-based Linux kernel module for application layer-based policy enforcement between field and supervisory networks; shipped a real-time rule monitoring/configuration manager and integrated a high-speed encryption-decryption module to enhance the communication security on the supervisory network side.

**Indian Institute of Technology Roorkee (IIT Roorkee)** Uttarakhand, India
*BARC sponsored (DGFS) M. Tech Fellow* *2015-2017*

- **Research Experience — Network Security**
  - **Design and Implementation of Remote Attestation protocols for servers:** Proposed and implemented a novel remote attestation protocol utilizing the system's underlying hardware and software measurements that successfully addressed the existing low extensibility and openness problems. [Published in IEEE TrustCom 2017]

## Publications

- **Monika Santra**, Bokai Zhang, Mark Lim, Vishnu Asutosh Dasu, Dongrui Zeng, Gang Tan. "iResolveX: Multi-Layered Indirect Call Resolution via Static Reasoning and Learning-Augmented Refinement." *NDSS*, under review, 2026.

- Bokai Zhang, **Monika Santra**, Shakthi Hussain, Gang Tan. "BPA-X: An Architecture-Agnostic Block-Based Points-to Analysis for Stripped Binaries." *NDSS*, 2026.

- Seok Hwan Kim, Dongrui Zeng, **Monika Santra**, Gang Tan. "BinType: Type-Based Indirect Call Target Refinement on Binary Programs." *ACM CODASPY*, 2026.

- Vishnu Asutosh Dasu, **Monika Santra**, Mohammad Rafid Ul Rashid, Ashish Kumar, Saeid Tizpaz-Niari, Gang Tan. "Heimdall: Formally Verified Automated Migration of Legacy eBPF Programs to Rust." *arXiv preprint*, 2026. arXiv:2605.25411

- Peicheng Wang, **Monika Santra**, Mingyu Liu, Cong Sun, Dongrui Zeng, Gang Tan. "DISA: Accurate Learning-based Static Disassembly with Attentions." *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*, 2025 (Cycle A), pp. 843–857. arXiv:2507.07246

- R. K. Koppanati, **M. Santra**, and S. K. Peddoju. "$D^2$4D: Dynamic Deep 4-Dimensional Analysis for Malware Detection." *IEEE Transactions on Information Forensics and Security (TIFS)*, vol. 20, pp. 2083–2095, 2025. doi: 10.1109/TIFS.2025.3531230

- **Monika Santra**. "AI-Augmented Static Analysis: Bridging Heuristics and Completeness for Practical Reverse Engineering." *ACM CCS Doctoral Symposium*, 2025.

- R. K. Koppanati, **M. Santra**, and S. K. Peddoju. "Defending AI/ML Models against Adversarial Attacks with Significant Contextual Features for Better Security." *Applied Soft Computing*, vol. 196, art. 115079, 2026. doi: 10.1016/j.asoc.2026.115079

- "**M. Santra**, S. K. Peddoju, A. K. Bhattacharjee and A. Khan. Design and Analysis of a Modified Remote Attestation Protocol.", 2017 IEEE Trustcom/BigDataSE/ICESS, Sydney, NSW, 2017, pp. 578-585

## Skills Summary

- **Computer Languages**: C, C++, Python, Java, Bash
- **Agentic AI / LLM**: LangChain, LangGraph, CrewAI, AutoGen, Google ADK, DSPy, Anthropic API, OpenAI API, prompt engineering, multi-agent orchestration
- **Packages:**

  GHIDRA, Angr, IDA pro, DWARF, pyelftools, capstone, TREX, Fairseq, Visual Studio Code, Qt Creator, Atom, Eclipse, Pycharm, Arduino Uno, VirtualBox, QEMU, KVM, Radare2, R2pipe, EMBER, SVF, Wireshark, Process Monitor, Kernel debugger, RPM package manager, Windows Driver Framework, Keras, weka, NumPy, Scikit-Learn, Matplotlib, SciPy, Pandas, AFL++, Pysa, PyTorch Geometric
- **Platforms**: Linux, Windows, Arduino
- **Soft Skills**: Academic Writing, Time Management, Endurance and perseverance towards research, Hardworking
- **Languages**: English, Hindi, and Bengali (Native Language)

## Research Interests

Binary analysis, malware analysis, security vulnerability analysis, cyber-resilient trusted & secure system design, system & software security, machine learning in security, agentic AI security

## Honors and Awards

- Recipient of ACM CCS Doctoral Symposium Travel Grant, *2025*
- Recipient of the Dr. Tse-Yun Feng Graduate Student Award (Best RA Award), *2023*
- Received ACM-W scholarship, *2017*
- Received Travel Grant Scheme from Indian National Academy of Engineering, *2017*
- I was the only computer engineer in the country in 2015 who received DAE Graduate Fellowship Scheme (DGFS-2015), Department of Atomic Energy, India, *2015-2017*
- Ranked among top 0.1% students all over the country in graduate test in engineering (GATE CS), *2015*

## SERVICE

- Reviewer: *IEEE Transactions on Network Science and Engineering (TNSE)*, 2024
- Reviewer: *Computer Networks*, 2025
- Poster Reviewer: *IEEE S&P*, 2025
- Organizer: Penn State INSR Industry Day, 2024, 2025